

Problem Solving Practice

HW Problem #1

CSE4152
Sogang University



Finding Celebrities

A celebrity is defined as a person who is known by everybody but does not know anyone. How can you find any celebrities among N people?

You can freely ask anyone as follows: "Hey, you know Mr. (or Ms.) X?"

Suppose that there are N people. Design a method that

- a) determines whether there is some celebrities among the people
- b) if there is any celebrities, find them efficiently (minimizing the number of questions you make).

Finding Celebrities – Guiding questions

- Can there be no celebrity at all?
- If a certain person X is considered a celebrity candidate, how can we verify whether X is actually a celebrity?
- Can there be exactly two celebrities?
- Can there be multiple celebrities?
- What can we learn from asking, “Does A know B ?”

2 Keys Keyboard

Initially, there is only one character 'A' on the screen. You can perform one of two operations:

- **Copy All:** copy all characters on the screen (partial copy is not allowed).
- **Paste:** paste the characters that were copied last time.

Given an integer N , return the minimum number of operations required to display exactly N 'A' characters on the screen.

Example

Example

- **Input:** $N = 3$
- **Output:** 3

Explanation:

- Step 1: Copy All $\rightarrow$ 'A' (clipboard = 'A')
- Step 2: Paste $\rightarrow$ 'AA'
- Step 3: Paste $\rightarrow$ 'AAA'

2 Keys Keyboard – Guiding questions

- Consider all the ways to make 8 A's.
- Consider all the ways to make 5 A's?
- Consider all the ways to make 6 A's?
 - Among these cases, which one is optimal?
- How can this idea be extended to N A's?

Minimum calculation

Imagine an electric calculator which has only two arithmetic operations, i.e., assignment operation($:=$) and multiplication operation ($*$). We want to let this machine compute $b = a^n$ for given numbers a and n .

Previously computed results can be stored in separate variables and reused in later multiplications. Design a method that produces a code for this machine, minimizing the number of multiplications.

Example when $n = 16$

$x_1 := a;$

$x_2 := x_1 * x_1; // a^2$

$x_3 := x_2 * x_2; // a^4$

$x_4 := x_3 * x_3; // a^8$

$x_5 := x_4 * x_4; // a^{16}$

$b := x_5$

Minimum Calculation – Guiding questions

- Consider the optimal way to compute a^{16} .
- Consider the optimal way to compute a^{15} .

Submission

- Please submit your report on the cyber campus.
- Please submit your report file using the following naming format:
 - CSE4152_STUDENTID_HW01.EXT (EXTENSION)
 - Ex) **CSE4152_20085934_HW01.doc / hwp / pdf**